

Blood Pressure Estimation Using Pulse Transit Time From Bioimpedance and Continuous Wave Radar

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Abstract— Objective: We have developed and tested a new architecture for pulse transit time (PTT) estimation at the central arteries using electrical bioimpedance, electrocardiogram, and continuous wave radar to estimate cuffless blood pressure. **Methods:** A transmitter and receiver antenna are placed at the sternum to acquire the arterial pulsation at the aortic arch. A four-electrode arrangement across the shoulders acquires arterial pulse across the carotid and subclavian arteries from bioimpedance as well as a bipolar lead I electrocardiogram. The PTT and pulse arrival times (PATs) are measured on six healthy male subjects during exercise on a bicycle ergometer. Using linear regression, the estimated PAT and PTT values are calibrated to the systolic and mean as well as diastolic blood pressure from an oscillometric device. **Results:** For all subjects, the Pearson correlation coefficients for PAT-SBP and PTT-SBP are -0.66 ($p = 0.001$) and -0.48 ($p = 0.0029$), respectively. Correlation coefficients for individual subjects ranged from -0.54 to -0.9 and -0.37 to -0.95 , respectively. **Conclusion:** The proposed system architecture is promising in estimating cuffless arterial blood pressure at the central, proximal arteries, which obey the Moens-Korteweg equation more closely when compared to peripheral arteries. **Significance:** An important advantage of PTT from the carotid and subclavian arteries is that the PTT over the central elastic arteries is measured instead of the peripheral arteries, which potentially reduces the changes in PTT due to vasomotion. Furthermore, the sensors can be completely hidden under a patients clothes, making them more acceptable by the patient for ambulatory monitoring.

Index Terms—Bioimpedance (Blmp), bistatic radar, continuous blood pressure, doppler radar, pulse arrival time (PAT), pulse transit time (PTT).

I. INTRODUCTION

ABNORMALLY high blood pressure (BP) or hypertension contributes to cardiovascular diseases (CVD) such as

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heart disease, stroke, and kidney failure which lead to premature death and disability. With one in three adults worldwide having high BP and more people dying from CVDs, hypertension is recognized by the World Health Organization as a global public health issue [1]. Since the first ambulatory BP measurements (ABPM) were carried out 50 years ago [2], strong evidence has emerged in several long-term studies that mean BP levels as well as BP variations over 24 h or longer are more sensitive and reliable predictors of cardiovascular risks than conventional measurements in the presence of a physician [3].

A method for estimating BP without a cuff is based on pulse transit time (PTT) or pulse arrival time (PAT), which is related to pulse wave velocity (PWV). A pressure pulse is generated at the point of the aortic valve opening, which propagates along the entire arterial tree. PWV refers to the velocity of propagation of the pulse wave across the arterial tree [4]. Several monitors based on PWV or PTT have been proposed in the literature and summarized in [5]. Many of the systems are compact, wearable, and fulfill requirements on accuracy of BP estimation. With the exception of Puke's [6] and Sola's [7] systems, many of the systems rely on measurement of the arterial pulse at the radial artery or the finger with the vasomotion being accounted for [8]–[10]. The peripheral arteries undergo vasomotion by activation of the smooth muscle, which changes the PWV independently of BP [11]. This factor reduces the validity of prior BP-PTT calibrations, forcing recalibrations to a cuff-based device as frequently as 6 min [12]. For clinical accuracy, measurement of the pulse wave signal at the central elastic arteries is therefore preferred [5], [11], [13].

The neck contains the carotid arteries, which are the possible locations to measure the PTT proximally. Piezoelectric sensors have been used since several years to obtain the carotid pulse [14]. Their use involves palpating or pressing against the neck to feel the mechanical pulse wave, which is uncomfortable and therefore, not feasible for ambulatory monitoring. Hsu and Young have reported a piezoresistive MEMS sensor capable of measuring the carotid pulse [15]. The sensor is packaged in a soft and hard silicone coating. It is not clear how much the neck must be palpated to acquire the pulse. A commercial accelerometer encapsulated in a custom-made housing is used to measure the pulse wave as well [16]. Although very compact and comfortable, the orientation of the sensor on the neck may results in the Ballistocardiogram appearing as an unwanted signal. A magnetic sensor has been reported in [17], where two

magnetoresistive sensors are placed on either side of a permanent magnet. Blood arterial pulsations influence the magnetic field, changing the resistance of the sensors. Although the signal quality in the aforementioned references is excellent, it is unclear to us how bulky the sensors will be or how much palpation will be required. Recently, a continuous wave (CW) radar sensor was proposed to measure the carotid pulse at the left common carotid artery without palpation [18]. This approach requires the antennas to be placed at the neck, which may not be favored by some patients in a clinical application.

In this investigation, we propose a method for estimating systolic BP (SBP) using CW radar as well as the electrocardiogram (ECG) and bioimpedance (BImp) across the shoulder. To the best of our knowledge, this method is the most compact architecture that measures the pulse wave signal, therefore estimating the PAT directly at the central arteries. The sensors can also be completely hidden underneath the clothes, enabling patient privacy. The BImp signal is hypothesized to acquire the pulse wave signal across the common carotid arteries. These are not expected to be affected by the smooth muscle contraction and surrounding tissue as the peripheral arteries would be [11]. By placing the CW radar antenna on the sternum, the CW radar signal is hypothesized to acquire the arterial distension of the ventral side of the aortic arch. Signal processing from existing literature is used to estimate the PTT from the aortic arch to the common carotid arteries, which is used to estimate BP from linear regression with a reference oscillometric device.

II. SYSTEM ARCHITECTURE

A. Overview

Fig. 1(A) shows a setup of three sensors on the shoulder and thorax: CW radar antennas, ECG, and BImp. The radar antennas are placed at positions 1a and 1b, which represent the transmitter and receiver antenna, respectively. Sensors 2a and 2b are BImp current injection electrodes, 3a and 3b are BImp and ECG measurement electrodes.

The hypothesis is that the BImp signal captured in Fig. 1(B) (second plot) is the impedance measured across the common carotid and subclavian arteries. The PAT is, therefore, the time difference between the ECG R -peak t_R and the rising slope t_2 of the BImp signal. The PTT is estimated in order to remove the preejection period (PEP) from the PAT. The PEP can change in opposite directions to the PTT during BP changes [19] due to changes in preload and afterload. For this purpose, the last local maximum or minimum before the largest change in the radar signal t_1 is identified as the point where the arterial pulse is propagating along the aortic arch.

The electrode arrangement shown in Fig. 1(C) and (D) indicates that the BImp current could possibly travel across the left and right subclavian arteries, which are displayed behind the four electrodes. Individual sensors are described in greater detail in the following sections. The signal processing is described in Section II-E.

B. CW Radar

1) Background: A CW radar system (see Fig. 2) will emit a narrow-band wave signal $T_{CW}(t)$ of radial frequency ω and

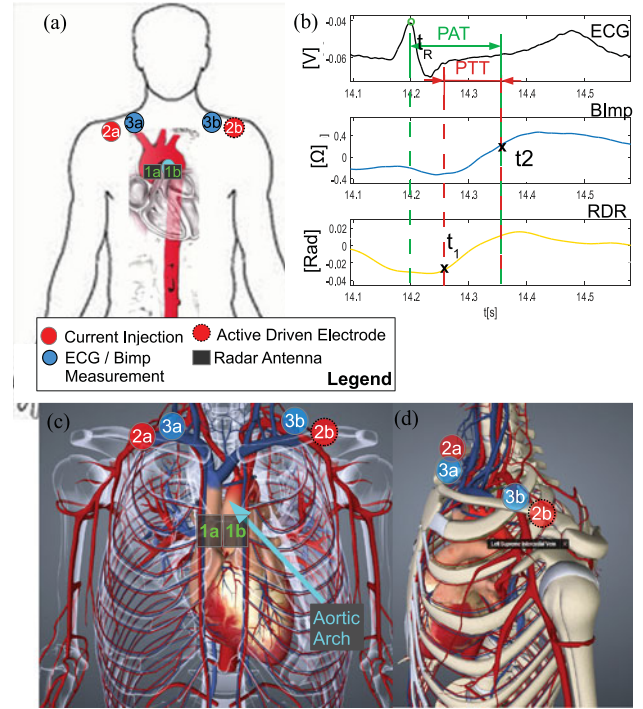


Fig. 1. (A) Placement of sensors. (B) Acquired signals from sensors with timings acquired at time stamps labeled “x.” BImp signal is inverted. (C) Anatomy of cardiovascular system (frontal view). (D) Partially lateral view. Image source for (C) and (D): human.biomedical.com.

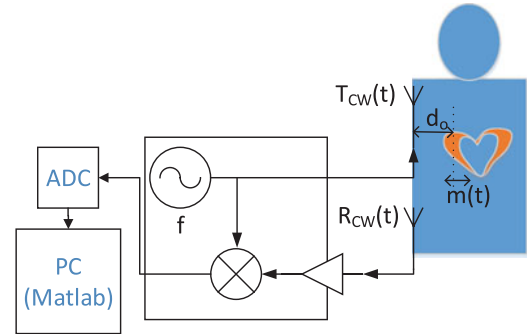


Fig. 2. Basic block diagram of a bistatic CW radar measuring motion of the heart. For brevity, only the in-phase mixer is shown.

phase noise ϕ_t [20]. If the target exhibits a periodic motion $m(t)$, e.g., artery or heart at a distance d_0 from the radar, the phase of the reflected wave $R_{CW}(t)$ will contain $m(t)$. For brevity in the analysis, we assume a homogenous layer of tissue in the thorax. Neglecting path loss and amplification before the mixer, the received signal $R_{CW}(t)$ will possess some properties of the transmitted signal:

$$T_{CW}(t) = E_o \cos(\omega t + \phi(t)) \quad (1)$$

$$R_{CW}(t) = E_1 \cos\left(\omega t + \phi_t(t - 2d_0/c) - \frac{4\pi d_0}{\lambda} - \frac{4\pi m(t)}{\lambda} + \theta_o\right). \quad (2)$$

Amplification, filtering, and demodulation of $R_{CW}(t)$ with a homodyne quadrature mixer will yield in-phase and quadrature baseband signals. Neglecting amplitude values, these signals are

expressed as:

$$I(t) = \cos \left(\underbrace{\phi_t - \phi_t(t - d_0/c)}_{\text{residual phase noise}} + \theta_o + \frac{4\pi d_o}{\lambda} + \underbrace{\frac{4\pi m(t)}{\lambda}}_{\text{desired phase}} \right) \quad (3)$$

$$Q(t) = \sin \left(\phi_t - \phi_t(t - d_0/c) + \theta_o + \frac{4\pi d_o}{\lambda} + \frac{4\pi m(t)}{\lambda} \right). \quad (4)$$

Signals $I(t)$ and $Q(t)$ are nonlinear functions of the desired phase. Several methods of phase recovery are reviewed in [21] for free space. One method of phase recovery that can work for small or large displacements [21] is the arctangent calculation: $\text{RDR}(t) = \arctan(\frac{Q(t)}{I(t)})$. Using on-body radar on the thorax, the multiple tissue layers act as scatterers, however our visual comparison between $\text{RDR}(t)$ (see Fig. 1) and M-mode echocardiographs from previous works [22], [23] shows many similarities between these signals. At 2.45 GHz, the wavelengths resulting from this carrier frequency as well as the thorax dielectric properties are in the order of a few cm [24]. This distance is in the same order as an example value of the distance between the anterior border of the aorta and the sternum (2.5 cm) [25], meaning that the aorta is definitely in the near field of the transmitter antenna.

One limitation of narrow-band radar is its vulnerability to external interferers. Ultrawideband (UWB) may be more suitable in overcoming this problem. However, the aorta is located a few cm away from the radar behind the lungs, muscle, fat, and skin. This represents a multilayered dielectric model in the antenna's near field, making signal processing of the received UWB signal very complex as compared to the narrow-band signal. Narrow-band is, therefore, chosen as a starting point.

2) Previous Literature: Most of the radar-based monitors have been studied in a contactless setting, where free space separates the antenna and the body [21], [26]. The dominating source of phase modulations has been proven to be the motion at the surface of the body [27]. When the radar antennas have been placed in contact with the body, the radar phase modulations have been hypothesized to be due to propagation path loss [28] within the body, body surface motions [29], [30], or changes in near-field coupling due to body tissue movement [24], [31], [32], or far-field motions of the heart [33].

The distance between the radar and the heart with respect to the carrier wavelength affects signal linearity and, hence, morphology [31], making feature-based identification of systolic and diastolic intervals difficult. The arctangent was not calculated in either publication, making signal processing of nonlinear signals impossible. Nonetheless, the authors' approach was interesting in the sense that they assume points of zero velocity in cardiac motion, which will appear as zero crossings in the time derivative of the radar signal [31], [34].

3) Antenna: Using ECG electrodes as adhesive stickers, two monopole patch antennas are mounted in a bistatic configuration on the sternum at the third intercostal space [see Fig. 1(A) and (C)]. The antennas have been designed previously for UWB propagation in a free space between 3.1 and 12 GHz (S_{11} coarse

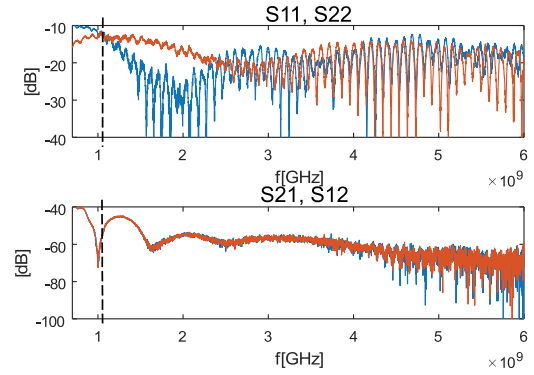


Fig. 3. S parameters for bistatic antenna arrangement. The vertical dashed black lines show the S values at the operating frequency of 1.1 GHz.

TABLE I
ELECTRICAL PROPERTIES OF TYPICAL HUMAN TISSUES AT 1.1 GHz [36]

Tissue Layer	Conductivity σ [S]	Permittivity ϵ_r	Wavelength λ [cm]
Skin	0.93	40.5	4.2
Fat	0.056	5.4	11.6
Bone	0.17	20.4	6.0
Lung (infl.)	0.49	21.7	5.8
Lung (defl.)	0.94	50.8	3.8
Aorta	0.76	44.4	4.1
Heart Muscle	1.34	58.8	3.5

dashed curve in Fig. 3). The body acts as a superstrate, which will influence the impedance seen at the antenna input [35]. Since the dielectric values of the body tissues are larger than that of free space, the operating frequency of the antenna will be smaller than its intended value in free space. This is shown in the S parameter measurements in Fig. 3. In order to investigate the proof of the concept of our proposed system architecture, the carrier frequency of the CW radar is selected because of the operating frequency of the antennas. The antennas were separated by three vertically stacked strips of RF absorbing material (TDK IRJ04AB 300X200X0.25) of dimensions 0.5 cm width, 8 cm length, and 0.25 mm thickness. The stackup resulted in a thickness of 0.75 mm. Furthermore, in order to direct the radiation toward the heart, the antenna PCBs were embedded into three stacked strips of the same RF absorbing material of 10 cm length and 5 cm width. By embedding the antennas in the absorbing material, most of the energy was transmitted either into the chest or absorbed in the absorbers, limiting the disturbance from other movements in the room [33].

The CW radar antennas are placed in front of the ascending aorta or the aortic arch. This may not be a realistic assumption as the heart's size and orientation within the thorax can vary across individuals. As seen in Fig. 1(C) and (D), the aortic arch is surrounded by the vena cava on the left and the pulmonary artery on the right and the right ventricle from below. The reflected RF wave is likely to contain phase modulations from these blood vessels as well making feature detection difficult.

Table I contains the electrical properties of human tissue at 1.1 GHz. Depending on the tissue layer that the transmitted

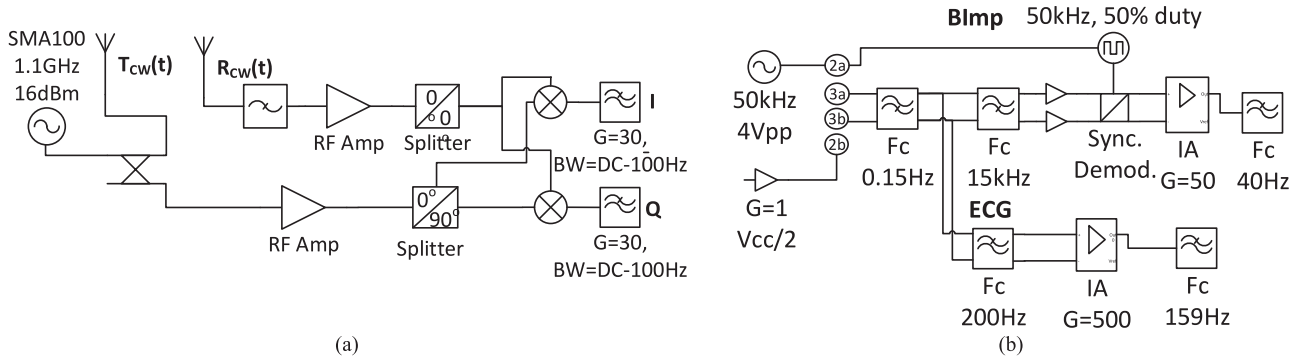


Fig. 4. Block diagram of readouts: (A) RF transceiver; (B) Blmp and ECG readout circuit.

CW signal passes through, the wavelength will be 4.1–11.6 cm. With the heart and ascending aorta being about 3 cm away from the antenna [33], a near-field scenario may be occurring. The received signal $R_{CW}(t)$ at the antenna will be affected by nearby reflective boundaries, resulting in $RDR(t)$ to vary in morphology between individuals. In one subject, $RDR(t)$ has also varied at different BP levels, which should be taken into account for feature detection when estimating t_1 .

The S_{11} parameter is the ratio of the reflected power to the fed power at the transmitter antenna input. Similarly, the S_{22} parameter measures the same for the receiver antenna. For proper antenna operation, these parameters should be less than -10 dB. The measured S_{11} and S_{22} parameters were below -15 dB at 1.1 GHz for all subjects. The emitted power fed to the antenna is 50 mW, which is less than the typical maximum power of 0.1–2 W emitted by cellphones. For any antenna used on or near the body, specified safety threshold values of the specific absorption rate (SAR, 1.6 W/kg for 1 g of tissue or 2.0 W/kg averaged over 10 g of tissue) must be obeyed [37], [38]. SAR estimation requires validation in simulation as well as measurement and is nontrivial. This investigation has focused on antenna positioning and signal processing to assist PTT estimation.

4) Radar System: The system is shown in Fig. 4(A). A signal generator (Rohde and Schwarz SMA 100 A) is used to generate a CW at 1.1 GHz for the transmitter antenna as well as the local oscillator (LO). This frequency is chosen as a tradeoff of all values of the S parameters for all subjects. While S_{11} and S_{22} values of less than -10 dB indicate functional transmitter and receiver antennas, low values of S_{21} and S_{12} will indicate that the near-field coupling will be as small as possible. A low value of all S parameters is, therefore, desired. Broadband RF amplifiers are used to amplify the received and local signals. The LO signal and received signal $R(t)$ are split using a quadrature and in-phase splitter, respectively. A double balanced mixer is used to produce the baseband signals $I(t)$ and $Q(t)$. The baseband signals are amplified and low-pass filtered using commercial chip amplifiers (OPA350 Texas Instruments).

C. Arterial Signal Using Bioimpedance

Measuring the arterial pulse makes use of plethysmographic-related impedance changes in the artery as well as significantly

smaller changes in resistivity of blood [39]. This would mean that impedance during systole would be smaller than that during diastole. However, for our investigation, the Blmp signal is always shown inverted. As seen in Fig. 1, a four-electrode arrangement is used, with one current injection electrode 2 and voltage measurement electrode 3 on either side of the neck. A four electrode measurement in Blmp means that the impedance between the two voltage measurement electrodes is measured, i.e., 3a and 3b [see Fig. 1(B) and (C)]. By limiting the positions of these to the neck, it is very likely that a combination of the carotid and subclavian arterial pulses is measured. Placing electrodes 2a–3b closer together around the neck is likely to limit the current pathway across the left and right common carotid arteries and, hence, limit the PTT measurement to these. However, the ECG signal measured between 3a and 3b will be smaller in amplitude, hence, the electrode distance between 3a and 3b is a tradeoff between the ECG R -peak amplitude and the origin of the measured Blmp signal variations.

A block diagram of the readout is shown in Fig. 4(B). A constant ac voltage was injected instead of a constant current. An ac-coupled sine voltage signal of $3V_{pp}$ at 50 kHz was injected at electrode 2a, while electrode 2b was kept a constant, buffered voltage of 2.5 V. A common differential high-pass filter at 0.5 Hz corner frequency is used to improve the common mode rejection [40]. A second differential high-pass filter of corner frequency 15 kHz is used to suppress the ECG signal between 0.1 and 100 Hz. The resulting voltage carrier signal is amplified differentially. The carrier signal is demodulated synchronously by a clock signal of 50% duty cycle and zero phase difference with respect to the sine wave [41]. The baseband signal is then amplified and converted to a single ended signal using a 3-opamp instrument amplifier (INA333), followed by an antialiasing low-pass RC filter.

D. ECG

The ECG signal is measured between electrodes 3a and 3b and further amplified by the readout in Fig. 4 B. The measured lead is similar to a bipolar Lead I configuration. Depending on the QRS axis of the heart, the QRS wave may be mono- or biphasic. Any QRS algorithm should either be robust to the QRS axis or be calibrated taking this into account.

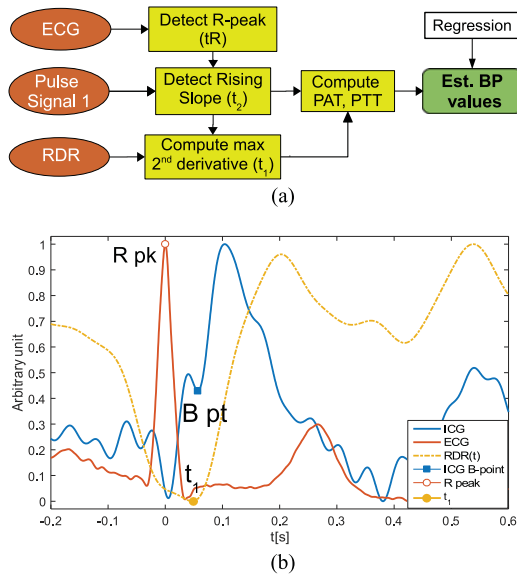


Fig. 5. (A) Signals (oval boxes) and signal processing steps (rectangular boxes) used to estimate BP values. (B) Comparison of t_1 of $RDR(t)$ with the R -peak and B-point of a simultaneously measured ECG and impedance cardiogram in one cardiac cycle. ICG spot electrode configuration from [44] was used.

E. Signal Processing

Three algorithms from the literature (see block diagram shown in Fig. 5) are used to detect the timings of the features of the three acquired signals as in Fig. 1(B). The timings are described in (7)–(9). The ECG R -peak at t_R is detected using the Pan–Tompkins algorithm [42]. The arrival of the BImp pulse wave (demarcated by t_2 , “x” in middle plot) is calculated using an tangent hyperbole function based on the method reported in [43]. The BImp signal is normalized with respect to its maximum amplitude and its arrival is detected using the minimum least squares difference between its rising slope and the arctangent function [43] $\arctan(\frac{t-t_s}{c})$, where c and t_s change the slope and center of the arctangent function. As seen in Fig. 1(B), the PAT is $t_2 - t_R$.

The radar signal is then used to assist in PTT estimation. The location of t_1 is hypothesized to be the diastolic foot at the aorta and is calculated using the maximum of the second derivative of $RDR(t)$. We estimate the value of t_1 assuming maximum distension at zero velocity of the hypothesized target organ (ascending aorta). In a proof of concept measurement of t_1 , we compare the radar signal to a simultaneously measured impedance cardiogram (ICG) using the spot electrode configuration from [44]. By proper positioning of the antenna in one subject, a large distension is shown to precede the B-points of the ICG by less than 10 ms [see Fig. 5(B)]. The B-points have been known to overestimate the opening of the aortic valve with respect to gold standard Doppler echocardiographic measurements in supine and tilt [45] by 12.6 ± 16.1 ms in tilted subjects. Using this rationale to select t_1 , we assume a zero crossing in the 1st derivative of $RDR(t)$ [24]. The location of t_1 [“o” in the top plot in Fig. 1(B)] must be in between t_R and t_2 . Once the value of t_1 is found, PTT is estimated by subtracting t_1 from t_2 .

In this study, the SBP (SBP_{est}) values from PAT/PTT are estimated using linear regression coefficients a and b (7–9).

$$PAT = t_2 - t_R \quad (5)$$

$$PTT = t_2 - t_1 \quad (6)$$

$$SBP_{est} = PAT * a_1 + a_2 \quad (7)$$

$$SBP_{est} = PTT * b_1 + b_2 \quad (8)$$

$$MBP_{est} = PTT * c_1 + c_2. \quad (9)$$

III. EXPERIMENTAL DESIGN

The experimental setup consists of the sensor placement and the hardware is shown in Fig. 6. All sensors are mounted using sticky biopotential electrodes of 6 cm diameter (3M Red Dot 2249). On six male subjects with no prior cardiovascular problems, we have measured the PTT and reference BP values before and after exercise on a bicycle ergometer. An isolation transformer was used to disconnect the mains from the patient. The data acquisition card was connected to a battery-operated laptop. The BImp root mean square current was measured using a Fluke 87V multimeter to ensure that the injected ac current was less than 0.1 mA according to the IEC-60601-2 standard.

A. Validation Experiments

One subject was made to lie in supine position. For estimation of t_2 , the antennas were placed at the neck at the left common carotid artery in a similar manner as our previous investigation [18]. This was done to validate whether the measured BImp signal is likely to contain components of the carotid arterial pulse. If this were to be the case, the arterial signal from the neck will always follow that of the BImp signal, i.e., its PAT value will be larger than t_2 , regardless of the PAT algorithm that is used.

In the second validation experiment, the goal was to validate the behavior of $RDR(t)$ during respiration as well as justify the use of t_1 as a feature. For the estimation of t_1 , the antennas were placed on the sternum according to Fig. 1. The subject was made to breath at 7 cycles/min. The first derivative of the BImp signal from the same hardware setup as shown in Fig. 4(B) was used to acquire the ICG. The spot electrode configuration of Sherwood *et al.* [44] was used.

B. BP Estimation

In order to vary the BP due to physical load, the experimental protocol was as follows:

- (1) Measure all signals for one minute as well as reference BP at “rest.” Conduct two measurements with a rest interval of 2.5 min to allow brachial artery to rest. The subject’s respiration rate was uncontrolled.
- (2) 2.5 min cycling at a speed of 15–18 km/h.
- (3) Measure signals at “moderate” BP as per item 1.
- (4) 2.5 min cycling at a speed of 25–30 km/h.
- (5) Measure signals at “high” BP as per item 1.

The analysis of the signals is done as follows. The signals are high- and low-pass filtered using finite-impulse response filters

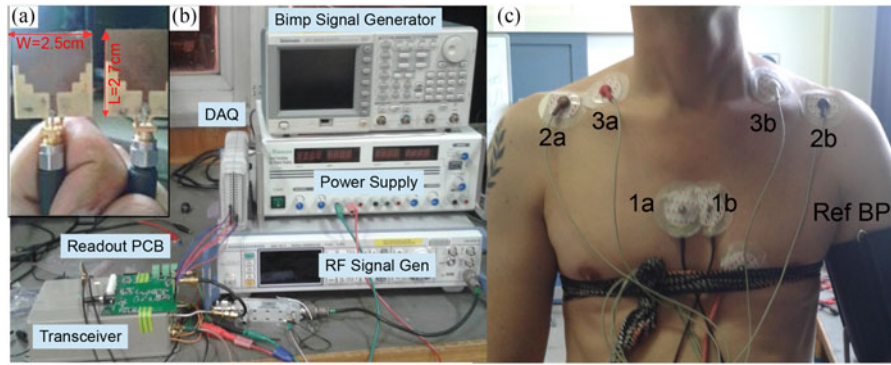


Fig. 6. Experimental setup. (A) Antennas, (B) hardware setup, and (C) sensor placement on human body, based on Fig. 1.

TABLE II
COMPARISON OF PAT VALUES FROM ALL SUBJECTS WITH LITERATURE
TAKEN AT RESTING BP

Artery Location	Signal feature	Posture	Median value [ms]
Left car. [46]	Foot	Supine, hypertensive patients	90
Both common car./subclavian (This work)	Rising slope	Sitting	105
Common car. [47]	Foot	Sitting	126
Left car. [18]	Rising slope	Sitting	133
Common car. [48]	Signal peak	Sitting	150

at 0.3 and 400 Hz, respectively. Using the Pan–Tompkins algorithm [42], ensemble averaging of all signals over four cardiac beats is carried out to increase SNR. For each 1-min recording, the mean PAT and PTT were computed from all ensemble averages. From the total of six measurements per subject, Pearson correlation coefficients between the BP values and PAT, PTT values were calculated for each subject as well as all subjects. A linear regression using Matlab’s polyfit function was carried out between PAT–SBP and PTT–SBP.

IV. RESULTS

A. Validation of Arrival Timings

1) Pulse arrival time t_2 : In order to understand whether the BImp signals are likely to be measuring across the carotid arteries, a comparison is done with previous literature in Table II. The results are sorted according to increasing values of the obtained PAT. The same feature (rising slope) was detected from the signals obtained in this study as well as a previous work [18], where the signals were obtained at the left common carotid arteries at the neck. Furthermore, the subject type and position were the same.

As a second piece of evidence, we have made additional measurements on the subjects in supine position (20 s/subject). The inverted BImp signal was measured with the radar placed at the neck [18]. Signals from one subject are given in Fig. 7(A) with the PAT values from three different algorithms in 7(B).

It can be seen visually that systolic upstroke of the carotid pulse using radar (dashed line) follows that of the BImp.

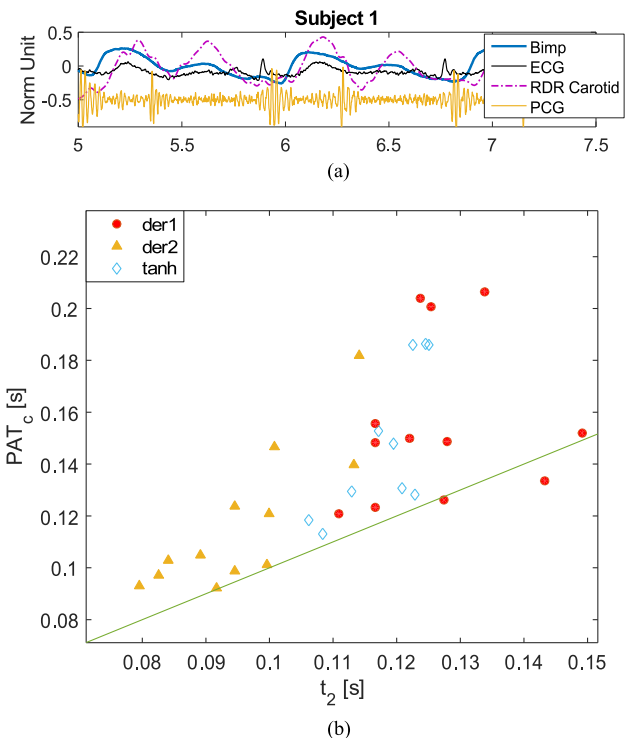


Fig. 7. (A) Signals from one subject with ECG, inverted BImp, and Carotid pulse using CW Radar. (B) PAT using three algorithms.

Furthermore, both the BImp and carotid signals contain a local minimum, which is just after the S2 heart sound. This coincides with the dichrotic notch, which is associated with aortic valve closure [14], [49]. As a further evidence using ensemble averaging over four beats, the PAT was calculated using three methods: maximum of the second derivative [50], first derivative [51], and best-fit with a tanh function [43]. The scatter plot of the PAT of carotid (y-axis) versus BImp (x-axis) is shown for all the three algorithms.

The straight green line indicates $PAT_c = t_2$. Ideally, PAT_c should be larger than t_2 for all algorithms. It can be seen that most of the PATs are higher for the neck using radar, as compared to the shoulder using BImp. The subtraction of the BImp PAT (derivative 2 method) from the neck radar PAT gives a median value of 33 ms and a mean absolute deviation of 26 ms.

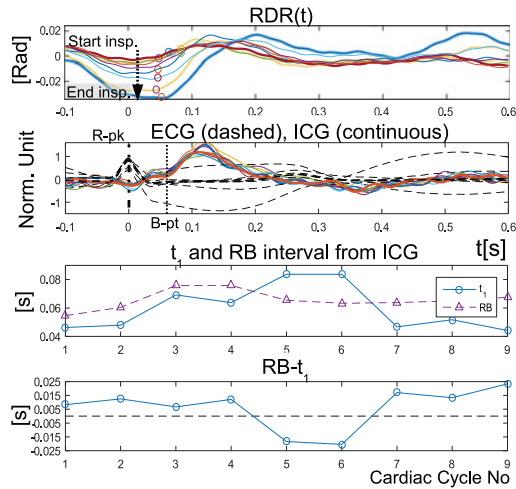


Fig. 8. Plot showing $RDR(t)$ over one respiration cycle. The signal at the beginning and end of inspiration is drawn with a thicker line.

Taking a distance of approximately 15-cm horizontal distance between the voltage measurement electrodes and the radar antenna, the PTT values translate into a PWV value of approximately $0.15 \text{ m}/(33 \text{ ms} - 26 \text{ ms}) = \text{median of } 4.55 \text{ m/s}$ and ranging between 2.53 and 22.7 m/s. The median PWV value is well within the expected range of 4–5 m/s at a supine position for the ascending aorta/carotid artery [4], [52].

2) Estimation of t_1 using radar: The top plot of Fig. 8 shows the $RDR(t)$ signals from several cardiac cycles over one inspiration cycle, i.e., from deflated to inflated lung. The second plot shows the ECG and ICG from the corresponding cardiac cycles. All the three signals are aligned to the ECG R -peak. The third plot shows the values of the RB interval from the ICG signal as well as t_1 from $RDR(t)$ for all nine cardiac cycles from a deflated to inflated lung. The B-point was calculated using the visually identified local maximum of the second derivative of the ICG, which occurred before the systolic upstroke. The last plot shows the value of $RB - t_1$.

It can be observed in the top plot that the amplitude of $RDR(t)$ increases with increasing lung volume. Given that the inflated lung has a similar permittivity to bone, as compared to the deflated lung (see Table I), a significantly larger proportion of the RF signal could be reaching the aorta/heart. Furthermore, by taping the antenna to the chest, we have observed a stronger physical coupling of the antenna to the skin on an inflated chest.

The third plot shows the values of both the RB interval from the ICG B-point and t_1 . The RB interval is used in this investigation as a surrogate for the PEP. Given that the subject was in supine position, the RB interval of 50–60 ms does not include the QR interval and is, therefore, in reasonable agreement with the PEP values of $95 \pm 19 \text{ ms}$ from supine healthy people [53]. The fourth plot shows that t_1 precedes the RB interval for most cardiac cycles by 5–15 ms for six out of nine cardiac cycles. These differences are well within the range of the over-estimation of the PEP from ICG as compared to the Doppler echocardiography [45].

There is a sudden increase in t_1 from cardiac cycle 4 to 5. This was due to the elementary derivative based that was

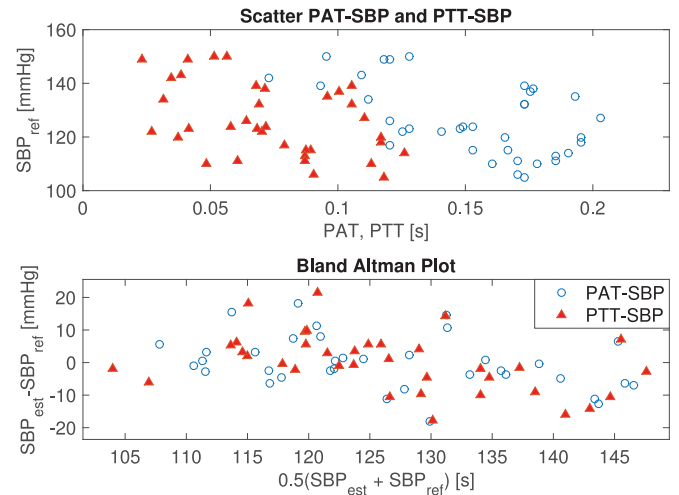


Fig. 9. (A) Scatter plot of reference SBP–PAT and reference SBP–PTT. (B) Bland Altman plots using SBP estimated from PAT and PTT.

used to calculate t_1 . However, for the nine cardiac cycles, the median difference between the RB and t_1 is approximately 15 ms. Another interesting observation is that the RB interval increases by as much as 20 ms from cardiac cycle 1–5. It has been suggested that an increase in PEP is correlated with an increase in the end-inspiration lung volume [54], which was the case in our measurements.

B. BP Estimation

The main objective of this investigation was the feasibility of the proposed architecture in Fig. 1 to estimate SBP from PTT. Fig. 9 provides an overview of the results. The top plot (A) shows a scatter plot of PAT–SBP as well as PTT–SBP pairs, respectively. Individual Pearson correlation coefficients are shown in Table III. For all the six subjects, the PTT values ranged from 0.025 to 0.128 s while PAT values ranged from 0.64 to 0.205 s. The Pearson correlation coefficient for both PTT–DBP and PTT–MBP is -0.4 with a p -value of 0.025 and 0.029, respectively, implying a correlation significantly different from zero. The correlation coefficient of SBP–PAT is -0.35 with a p -value of 0.056, implying a weaker significance of the correlation. For each subject, the estimated SBP, MBP, and DBP were calculated using a linear regression coefficients from each individual subject. The estimated values are shown in a Bland Altman plot in Fig. 9(B).

The scatter plots of the PAT and PTT versus the reference SBP are shown in Fig. 10 and Table III, and Fig. 10 shows that there is moderate to strong correlation between PAT and SBP for all the subjects. Only subjects 1 and 5 continue to show a strong PTT–SBP correlation.

V. DISCUSSION

The architecture proposed in this system was expected to lead to a compact chest-based BP monitor when compared to the state of the art [6], [55]. The BImp signal produced in this investigation was hypothesized to be due to variations in the carotid artery. The PATs are similar to those in

TABLE III
PEARSON CORRELATION COEFFICIENTS FOR INDIVIDUAL SUBJECTS

	PAT-SBP r_1	PTT-SBP r_2	PAT-MBP r_3	PTT-MBP r_4	PAT-DBP r_5	PTT-DBP r_6
Overall	-0.66 (0.001)	-0.48 (0.0029)	-0.39/(0.038)	-0.49/(0.002)	-0.40/(0.04)	-0.33/(0.048)
Subj. 1	-0.9 (0.0155)	-0.8 (0.0584)	-0.84 (0.0368)	-0.58 (0.225)	-0.61 (0.197)	-0.21 (0.694)
Subj. 2	-0.55 (0.263)	-0.41 (0.414)	-0.7 (0.123)	-0.64 (0.171)	-0.69 (0.127)	-0.78 (0.07)
Subj. 3	-0.83 (0.0395)	-0.7 (0.118)	-0.71 (0.112)	-0.79 (0.0618)	-0.074 (0.889)	-0.52 (0.287)
Subj. 4	-0.74 (0.0354)	-0.57 (0.233)	-0.8 (0.0549)	-0.52 (0.287)	-0.59 (0.215)	-0.32 (0.536)
Subj. 5	-0.88 (0.0202)	-0.95 (0.00303)	-0.92 (0.00839)	-0.95 (0.00341)	-1 (2.88e-05)	-0.85 (0.0336)
Subj. 6	-0.54 (0.27)	-0.37 (0.47)	-0.45 (0.375)	-0.31 (0.545)	-0.36 (0.48)	-0.26 (0.617)

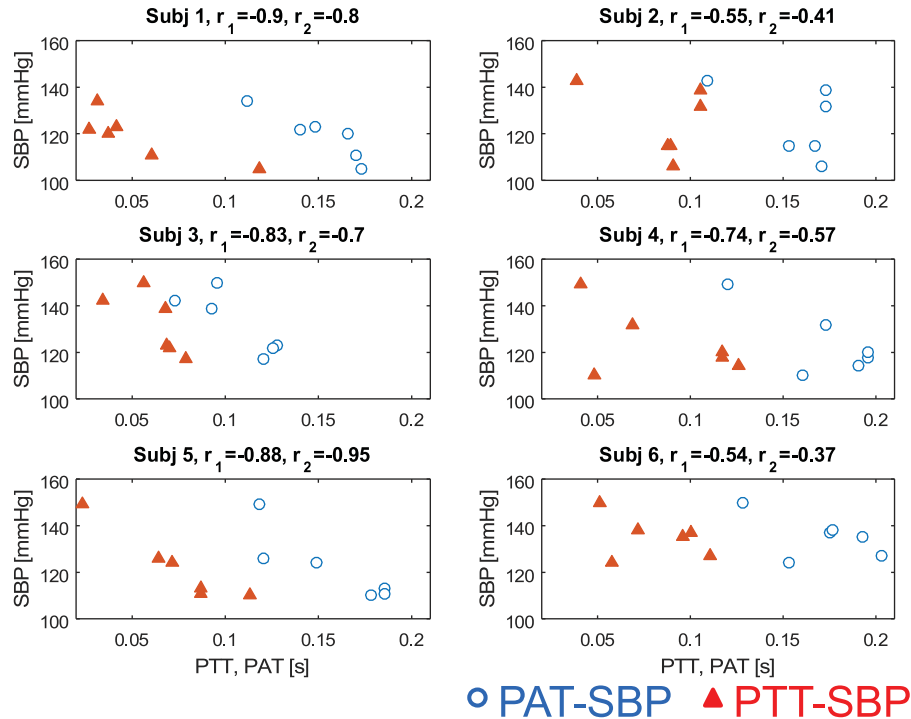


Fig. 10. PAT-SBP and PTT-SBP scatter plots of individual subjects. Values of r_1 and r_2 refer to Pearson correlation coefficients of PAT-SBP and PTT-SBP, respectively.

prior investigations in Table II as well as the internal thoracic artery, which has comparable distances to the aortic valve [55].

The BImp and ECG signals shown in Fig. 1(B) are reproducible across all the subjects and BP levels. The feet of the inverted BImp pulse wave signals measured across all subjects are somewhat more blunted than a typical photoplethysmograph at the radial artery or recently, the carotid arterial signal using CW radar [18]. This investigation raises the question as to which arteries are contributing to the PAT, e.g., common carotid and the subclavian arteries. Shortening of the electrode distance may restrict measurement to the common carotid arteries, but the ECG R -peak beat detection in noisy conditions may be compromised. The ECG lead used in this architecture is similar to bipolar lead I. We have also observed that the QRS wave can be more biphasic, depending on the heart's electrical axis. This may possibly make R -peak detection more difficult. The optimal electrode arrangement needs to be investigated further. Additionally, a more

robust algorithm such as Wavelet Transform [56] to work better as compared to Pan-Tompkins used in this investigation [42].

In contrast with the ECG and BImp signals, the radar signals are not as reproducible in shape across different subjects. This challenge has been stated in Thijs and Muehlsteff's investigations [31], although they worked with the I and Q signals instead calculating the arctangent of Q/I [57]. It is known that the heart size, position within the thorax, and orientation will vary with each individual. Although the sternum is attractive for antenna placement, given its relative flatness and small lung volume at that location, this investigation indicates that other locations on the chest should be investigated for producing a signal feature that is uniform in shape across different persons. Using an on-body antenna at 1 GHz with the heart/aorta approximately 2–5 cm away [58] means that these target organs are within in the antenna's near field. Given an inhomogenous and uneven contour of the reflection surfaces within the body, the reflection from the target and clutter will vary depending on the antenna's

location and BP. The result of these uneven and uncontrolled reflections is that the signal shape will change depending on the antenna and anatomic locations as well as BP. The radiation pattern of the antenna on the thorax must be studied further to improve its directivity in order to isolate movement from the target organ of interest. A methodology is also needed to position the antenna based on its electrical axis and size. These steps could increase the reproducibility of a useful signal feature regardless of subject or BP. It is possible that the radar antenna at the sternum may not be an optimal position for estimating the PEP. Proper antenna positioning should also take into account the female anatomy at the thorax. Finally, it is seen in Fig. 8 that the radar signal shape changes during the respiration cycle. This is most likely due to the antenna field of view changing with respect to the target organ. Moreover, changing lung volume will affect both the antenna's S_{11} and S_{22} parameters as well as the far-field propagation of the RF wave. Signal processing techniques are required to detect t_1 . Alternatives to using radar for this purpose are the ICG signal [55], or more recently, the Ballistocardiogram [59] or Seismocardiogram from accelerometers on the chest [60], [61].

The overall results from Fig. 9 and Table III indicate that the architecture proposed in this investigation has the potential to at least estimate SBP. Admittedly, the moderate values of the Pearson correlation coefficients of PAT (−0.66) and PTT (−0.48) versus SBP are lower than those of −0.73 to −0.99 from the literature, where PAT was correlated with SBP during exercise [62]–[66]. This has several reasons. The correlation constant between BP from oscillometric device and a median PAT/PTT value from a central pulse will have introduced errors. It is known that an oscillometric device overestimates SBP [67], whose difference with respect to central SBP will be even larger than the SBP measured from an auscultatory monitor. Second, the ECG/BImp circuit may not provide the best quality signal. Further circuit redesigns as well as ensemble averaging should improve the signal quality, leading to a more accurate PAT estimation. Third, there is an error in the estimation of t_1 using the derivative method. There is a direct RF leakage from the transmitter to the receiver antenna via the superficial tissue layers. This RF leakage will appear as a unwanted nonlinearity or high-frequency oscillations [20] in our radar signals above 10 Hz. The arctangent method to retrieve the radar phase requires that the direct leakage has been eliminated in the signal processing [68]. This requirement is not necessarily fulfilled when using body-contact antennas, indicating that other signal processing methods should be considered. The oscillations are amplified in the first and second derivatives of the radar signal. Low-pass filtering of these oscillations removes useful signal content in addition to these oscillations. Most of methods reported in [21] have been used for in contactless radar for heart rate and respiration monitoring. Using body-contact radar in our investigation for PTT monitoring is one of the first investigations in the literature. We hope that further research will trigger better antenna design and placement as well as signal processing to achieve higher correlations.

It is known from the literature that the PEP has been known to change due to preload and afterload changes, which do not

necessarily change linearly with the SBP [19]. The radar sensor was investigated for its ability to estimate the arrival time t_1 at the proximal part of the aorta, which would be used to eliminate the PEP from the PAT of t_2 . The PTT–SBP correlation coefficient of −0.49 in our study is similar to the mean value of −0.46 (finger PTT with PEP estimated from the onset of the S1 heart sound in supine subjects [62]). The comparable results suggest that the current method of estimating t_1 might indicate the end of the PEP instead of accurately estimating t_1 of a pulse wave arrival at the aorta. It may also be possible that the radar sensor may have captured right ventricular or atrial movement as opposed to the aortic one. Stronger correlations between the PTT and SBP can be achieved by improving the antenna design or using antenna arrays [69]. Additionally, antennas must be placed on the chest such that the phase modulations can be restricted to the aorta or left ventricle for supine, sitting, and standing body postures. Signal processing methods must be implemented to take into account antenna field of view at different phases of a respiration cycle [70]. A more robust method should be used instead of second derivative calculation. Finally, the accuracy of t_1 estimation must be compared with the Doppler or M-mode echocardiography. Our architecture must be studied with the PTT–BP mapping methods proposed in previous studies to understand the limitations of mapping the same PTT onto MBP and DBP. For this purpose, a wearable system must be developed and tested under BP-changing stressors other than exercise [11]. Where wearability of the sensors was concerned, positive feedback was received about patient privacy being enabled because all sensors could be placed completely under the clothes.

VI. CONCLUSION

In this study, we propose a novel method of obtaining PAT and PTT from the carotid and subclavian arteries and the sternum using four electrodes at the shoulder and radar antennas on the sternum. These arteries are considered by us to be proximal and central with respect to peripheral ones such as the radial artery. The proposed system architecture enables the BP monitor to function while being completely hidden under a patient's clothes. Using a bicycle ergometer, we conclude that moderate to strong correlations between PAT and SBP in all subjects, while moderate and strong correlations are seen between PTT and SBP in 3 and 2 subjects, respectively. A bistatic radar configuration at the sternum can assist in estimating SBP based solely on PTT. However, challenges remain in accurate estimation of t_1 , namely better antenna design and positioning and subsequent signal processing over a respiration cycle. The applications of this system architecture can range from ambulatory BP as well as a limited range of cardiac arrhythmia monitoring, e.g., extra systole, especially during sleep and sports such as cycling. Additional future work in CW radar includes the estimation of the SAR of the antennas, when placed on the sternum as well as antenna design for use in the industrial scientific and medical bands. The architecture must be studied using stressors such as hand grip, cold press, and mental arithmetic to investigate limitations of this system architecture.

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